DATA DICTIONARY

Web-Based ECU Tuning Service and Sales Management System with Payment and Sales Analytics for Ranao Moto

A. Users and Access

1. ROLES

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD001	Role	role_id	PK. Unique identifier of the role	N/a	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD002	Role	role_name	Name of the user role	Role	N/a	N/a	Y	Alphanumeric	50	Admin, Staff, Technician (others may be added)
DD003	Role	description	Description of what the role is allowed to do	N/a	N/a	N/a	N	Alphanumeric	255	Any; optional
DD004	Role	status	Current status of the record	Record status	N/a	N/a	N	Alphanumeric	8	Active, Inactive

2. USERS

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD005	User	user_id	PK. Unique identifier of the user	User ID	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD006	User	role_id	FK. Role assigned to the user	Role	Role	role_id	N	Integer (unsigned)	10	Must exist in ROLES
DD007	User	username	Login name of the user	Login name, Account name	N/a	N/a	Y	Alphanumeric	50	Letters, digits, and underscores only. Case sensitive.
DD008	User	password	Hashed login password of the user (never stored as plain text)	Passcode	N/a	N/a	N	Alphanumeric	255	Stored as a secure hash (e.g., bcrypt)
DD009	User	first_name	Given name of the user	Given name	N/a	N/a	N	Alphanumeric	100	Letters, spaces, hyphens, and periods only
DD010	User	last_name	Family name of the user	Surname	N/a	N/a	N	Alphanumeric	100	Letters, spaces, hyphens, and periods only
DD011	User	status	Whether the user account is active	Record status	N/a	N/a	N	Alphanumeric	8	Active, Inactive
DD012	User	created_at	Date and time the record was created	Date created	N/a	N/a	N	DateTime	N/a	Auto-generated
DD013	User	updated_at	Date and time the record was last modified	Date modified	N/a	N/a	N	DateTime	N/a	Auto-generated

3. ACTIVITY_LOGS

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD014	Activity Log	log_id	PK. Unique identifier of the activity log	Audit ID	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD015	Activity Log	user_id	FK. User who performed the action	Actor	User	user_id	N	Integer (unsigned)	10	Must exist in USERS
DD016	Activity Log	action	Action performed by the user	Event	N/a	N/a	N	Alphanumeric	100	e.g., Login, Create, Update, Delete
DD017	Activity Log	module	System module where the action happened	Screen, Feature	N/a	N/a	N	Alphanumeric	100	e.g., Sales, Payments, ECU Tuning, Users; optional
DD018	Activity Log	description	Details of the action performed	N/a	N/a	N/a	N	Alphanumeric	255	Any; optional
DD019	Activity Log	ip_address	IP address of the device used	IP	N/a	N/a	N	Alphanumeric	45	Valid IPv4 or IPv6 address; optional
DD020	Activity Log	created_at	Date and time the record was created	Date created	N/a	N/a	N	DateTime	N/a	Auto-generated

4. NOTIFICATIONS

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD021	Notification	notification_id	PK. Unique identifier of the notification	Alert ID	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD022	Notification	user_id	FK. User who receives the notification	Recipient	User	user_id	N	Integer (unsigned)	10	Must exist in USERS
DD023	Notification	notification_type	Category of the notification	Alert type	N/a	N/a	N	Alphanumeric	50	e.g., Payment, Tuning Update, System; optional
DD024	Notification	message	Text of the notification	Alert message	N/a	N/a	N	Alphanumeric	255	Any
DD025	Notification	created_at	Date and time the record was created	Date created	N/a	N/a	N	DateTime	N/a	Auto-generated

B. Customers and Motorcycles

5. ADDRESS

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD026	Address	address_id	PK. Unique identifier of the address	N/a	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD027	Address	barangay	Barangay where the customer is located	Brgy., Locality	N/a	N/a	N	Alphanumeric	100	Letters, digits, spaces, and periods only

6. CUSTOMERS

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD028	Customer	customer_id	PK. Unique identifier of the customer	Client ID	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD029	Customer	customer_name	Full name of the customer	Client name	N/a	N/a	N	Alphanumeric	150	Letters, spaces, hyphens, and periods only
DD030	Customer	contact_number	Mobile or phone number of the customer	Phone number, Mobile no.	N/a	N/a	N	Alphanumeric	20	Digits, plus sign (+), and hyphens only; optional
DD031	Customer	address_id	FK. Address of the customer	Customer address	Address	address_id	N	Integer (unsigned)	10	Must exist in ADDRESS; optional
DD032	Customer	status	Whether the customer record is active	Record status	N/a	N/a	N	Alphanumeric	8	Active, Inactive
DD033	Customer	created_at	Date and time the record was created	Date created	N/a	N/a	N	DateTime	N/a	Auto-generated
DD034	Customer	updated_at	Date and time the record was last modified	Date modified	N/a	N/a	N	DateTime	N/a	Auto-generated

7. MOTORCYCLE_BRAND

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD035	Motorcycle Brand	motorcycle_brand_id	PK. Unique identifier of the motorcycle brand	N/a	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD036	Motorcycle Brand	model	Brand and model name of the motorcycle	Make and model, Unit model	N/a	N/a	N	Alphanumeric	100	Letters, digits, spaces, and hyphens only
DD037	Motorcycle Brand	year	Model year of the motorcycle	Year model	N/a	N/a	N	Year	4	1901–2155; optional

8. MOTORCYCLES

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD038	Motorcycle	motorcycle_id	PK. Unique identifier of the motorcycle	Unit ID	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD039	Motorcycle	customer_id	FK. Customer who owns the motorcycle	Owner	Customer	customer_id	N	Integer (unsigned)	10	Must exist in CUSTOMERS
DD040	Motorcycle	plate_number	Registration plate number of the motorcycle	Plate no., License plate	N/a	N/a	Y	Alphanumeric	20	Letters, digits, spaces, and hyphens only
DD041	Motorcycle	motorcycle_brand_id	FK. Brand and model of the motorcycle	Brand	Motorcycle Brand	motorcycle_brand_id	N	Integer (unsigned)	10	Must exist in MOTORCYCLE_BRAND
DD042	Motorcycle	color	Color of the motorcycle	Unit color	N/a	N/a	N	Alphanumeric	50	Letters and spaces only; optional
DD043	Motorcycle	status	Whether the motorcycle record is active	Record status	N/a	N/a	N	Alphanumeric	8	Active, Inactive
DD044	Motorcycle	created_at	Date and time the record was created	Date created	N/a	N/a	N	DateTime	N/a	Auto-generated

C. Services

9. CATEGORIES

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD045	Category	category_id	PK. Unique identifier of the category	N/a	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD046	Category	category_name	Name of the service category	Service type	N/a	N/a	Y	Alphanumeric	100	Letters, digits, and spaces only
DD047	Category	description	Description of the category	N/a	N/a	N/a	N	Alphanumeric	255	Any; optional
DD048	Category	status	Current status of the record	Record status	N/a	N/a	N	Alphanumeric	8	Active, Inactive

10. SERVICES

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD049	Service	service_id	PK. Unique identifier of the service	Service code	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD050	Service	category_id	FK. Category the service belongs to	Service type	Category	category_id	N	Integer (unsigned)	10	Must exist in CATEGORIES
DD051	Service	service_name	Name of the service offered (e.g., ECU tuning)	Service title	N/a	N/a	N	Alphanumeric	150	Letters, digits, spaces, and hyphens only
DD052	Service	service_price	Standard price of the service	Rate, Service fee	N/a	N/a	N	Currency	10,2	0.00–99,999,999.99
DD053	Service	description	Description of the service	N/a	N/a	N/a	N	Alphanumeric	255	Any; optional
DD054	Service	status	Whether the service is currently offered	Record status	N/a	N/a	N	Alphanumeric	8	Active, Inactive

D. Sales and ECU Tuning

11. SALES

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD055	Sale	sale_id	PK. Unique identifier of the sale	Sale no., Transaction ID	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD056	Sale	user_id	FK. Staff member who processed the sale	Cashier	User	user_id	N	Integer (unsigned)	10	Must exist in USERS
DD057	Sale	customer_id	FK. Customer who availed the service	Client	Customer	customer_id	N	Integer (unsigned)	10	Must exist in CUSTOMERS
DD058	Sale	sale_date	Date of the sale	Transaction date	N/a	N/a	N	Date	N/a	YYYY-MM-DD
DD059	Sale	total_amount	Total amount of the sale (sum of sale detail subtotals)	Grand total	N/a	N/a	N	Currency	12,2	0.00–9,999,999,999.99
DD060	Sale	status	Payment/processing status of the sale	Sale status	N/a	N/a	N	Alphanumeric	10	Pending, Paid, Partial, Cancelled
DD061	Sale	created_at	Date and time the record was created	Date created	N/a	N/a	N	DateTime	N/a	Auto-generated
DD062	Sale	updated_at	Date and time the record was last modified	Date modified	N/a	N/a	N	DateTime	N/a	Auto-generated

12. SALE_DETAILS

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD063	Sale Detail	sale_detail_id	PK. Unique identifier of the sale detail	Line item ID	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD064	Sale Detail	sale_id	FK. Sale the line item belongs to	Transaction	Sale	sale_id	N	Integer (unsigned)	10	Must exist in SALES
DD065	Sale Detail	service_id	FK. Service sold in the line item	N/a	Service	service_id	N	Integer (unsigned)	10	Must exist in SERVICES
DD066	Sale Detail	motorcycle_id	FK. Motorcycle the service was performed on	Unit	Motorcycle	motorcycle_id	N	Integer (unsigned)	10	Must exist in MOTORCYCLES
DD067	Sale Detail	quantity	Number of units of the service sold	Qty	N/a	N/a	N	Integer (unsigned)	10	1–999
DD068	Sale Detail	unit_price	Price per unit at the time of sale	Price each	N/a	N/a	N	Currency	10,2	0.00–99,999,999.99
DD069	Sale Detail	subtotal	Line total (quantity x unit price)	Line total	N/a	N/a	N	Currency	12,2	0.00–9,999,999,999.99

13. ECU_TUNING

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD070	ECU Tuning	tuning_id	PK. Unique identifier of the ECU tuning	Tuning job no.	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD071	ECU Tuning	sale_detail_id	FK. Sale line item this tuning job is billed under	N/a	Sale Detail	sale_detail_id	N	Integer (unsigned)	10	Must exist in SALE_DETAILS
DD072	ECU Tuning	motorcycle_id	FK. Motorcycle that was tuned	Unit	Motorcycle	motorcycle_id	N	Integer (unsigned)	10	Must exist in MOTORCYCLES
DD073	ECU Tuning	user_id	FK. Technician who performed the tuning	Technician	User	user_id	N	Integer (unsigned)	10	Must exist in USERS
DD074	ECU Tuning	tuning_date	Date the tuning was performed	Service date	N/a	N/a	N	Date	N/a	YYYY-MM-DD
DD075	ECU Tuning	ecu_type	Type or model of the ECU	ECU model	N/a	N/a	N	Alphanumeric	100	Letters, digits, spaces, and hyphens only; optional
DD076	ECU Tuning	ecu_before	ECU settings/map recorded before tuning	Stock map, Original settings	N/a	N/a	N	Text	65,535	Any; optional
DD077	ECU Tuning	ecu_after	ECU settings/map recorded after tuning	Tuned map, New settings	N/a	N/a	N	Text	65,535	Any; optional
DD078	ECU Tuning	tuning_status	Progress of the tuning job	Job status	N/a	N/a	N	Alphanumeric	12	Pending, In Progress, Completed, Cancelled

E. Purchases

14. PURCHASES

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD079	Purchase	purchase_id	PK. Unique identifier of the purchase	Purchase order no.	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD080	Purchase	user_id	FK. Staff member who recorded the purchase	N/a	User	user_id	N	Integer (unsigned)	10	Must exist in USERS
DD081	Purchase	purchase_date	Date of the purchase	Date bought	N/a	N/a	N	Date	N/a	YYYY-MM-DD
DD082	Purchase	reference_number	Supplier invoice or receipt number	Invoice no., Receipt no.	N/a	N/a	N	Alphanumeric	50	Letters, digits, and hyphens only; optional
DD083	Purchase	total_amount	Total amount of the purchase	Grand total	N/a	N/a	N	Currency	12,2	0.00–9,999,999,999.99
DD084	Purchase	status	Payment/processing status of the purchase	Purchase status	N/a	N/a	N	Alphanumeric	10	Pending, Paid, Partial, Cancelled
DD085	Purchase	created_at	Date and time the record was created	Date created	N/a	N/a	N	DateTime	N/a	Auto-generated
DD086	Purchase	updated_at	Date and time the record was last modified	Date modified	N/a	N/a	N	DateTime	N/a	Auto-generated

15. PURCHASE_DETAILS

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD087	Purchase Detail	purchase_detail_id	PK. Unique identifier of the purchase detail	N/a	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD088	Purchase Detail	purchase_id	FK. Purchase the line item belongs to	N/a	Purchase	purchase_id	N	Integer (unsigned)	10	Must exist in PURCHASES
DD089	Purchase Detail	quantity	Number of units purchased	Qty	N/a	N/a	N	Integer (unsigned)	10	1–9,999
DD090	Purchase Detail	unit_cost	Cost per unit	Cost each	N/a	N/a	N	Currency	10,2	0.00–99,999,999.99
DD091	Purchase Detail	subtotal	Line total (quantity x unit cost)	Line total	N/a	N/a	N	Currency	12,2	0.00–9,999,999,999.99

F. Payments

16. PAYMENT_METHOD

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD092	Payment Method	payment_method_id	PK. Unique identifier of the payment method	N/a	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD093	Payment Method	method_name	Name of the payment method	Mode of payment	N/a	N/a	Y	Alphanumeric	50	Cash, GCash, Bank Transfer (others may be added)
DD094	Payment Method	description	Description of the payment method	N/a	N/a	N/a	N	Alphanumeric	255	Any; optional
DD095	Payment Method	status	Current status of the record	Record status	N/a	N/a	N	Alphanumeric	8	Active, Inactive

17. PAYMENTS

ID	Business Data Object	Field Name	Description	Alternate Names	Associated Business Data Object	Data Field	Unique Value?	Data Type	Length	Valid Values
DD096	Payment	payment_id	PK. Unique identifier of the payment	Payment no.	N/a	N/a	Y	Integer (unsigned)	10	Auto-generated
DD097	Payment	sale_id	FK. Sale being paid (blank if the payment is for a purchase)	N/a	Sale	sale_id	N	Integer (unsigned)	10	Must exist in SALES; required if purchase_id is blank
DD098	Payment	purchase_id	FK. Purchase being paid (blank if the payment is for a sale)	N/a	Purchase	purchase_id	N	Integer (unsigned)	10	Must exist in PURCHASES; required if sale_id is blank
DD099	Payment	payment_method_id	FK. Method used to pay	Mode of payment	Payment Method	payment_method_id	N	Integer (unsigned)	10	Must exist in PAYMENT_METHOD
DD100	Payment	received_by	Name of the person who received the payment	Cashier name	N/a	N/a	N	Alphanumeric	100	Letters, spaces, and periods only; optional
DD101	Payment	amount_paid	Amount paid in this payment	Payment amount	N/a	N/a	N	Currency	12,2	0.00–9,999,999,999.99
DD102	Payment	reference_number	Transaction reference number (e.g., GCash or bank reference)	Ref. no.	N/a	N/a	N	Alphanumeric	50	Letters, digits, and hyphens only; optional
DD103	Payment	payment_status	Status of the payment	Payment state	N/a	N/a	N	Alphanumeric	10	Pending, Completed, Failed, Refunded
DD104	Payment	payment_date	Date and time the payment was made	Date paid	N/a	N/a	N	DateTime	N/a	Auto-generated